

Intelligent Reflecting Surface Assisted Anti-Jamming Communications in THz Wideband Systems: A Fuzzy WoLF-PHC Learning Approach with Hybrid Beamforming

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Abstract—Wideband terahertz (THz) communications offer multi-gigahertz spectrum access but are acutely vulnerable to targeted jamming owing to their narrow, highly directional beams. This paper proposes a unified framework for intelligent reflecting surface (IRS)-assisted anti-jamming in THz wideband systems, jointly optimizing transmit power allocation, IRS phase shifts, and hybrid analog-digital beamforming against a smart adaptive jammer. The resulting optimization problem is non-convex, high-dimensional, and subject to a dynamic adversary with unknown strategy; we decompose it via a three-block alternating optimization (AO) framework that solves each block in closed form or via reinforcement learning (RL). The novel fuzzy Win-or-Learn-Fast policy hill-climbing (WoLF-PHC) RL agent manages discrete power allocation using a 27-component fuzzy state abstraction and win/loss-adaptive learning rates, converging to a mixed strategy that is inherently unpredictable to a jammer monitoring action history. The THz system model incorporates $f_c = 100$ GHz, $B = 10$ GHz wideband OFDM (64 subcarriers), a 64-antenna sub-connected base station with $N_{\text{RF}} = 8$ RF chains, and a 256-element sub-array phase-dependent partial-connection (SPDP) RIS for wideband beam-squint compensation. Evaluated over five independent seeds with 600 training episodes each, the proposed method achieves 11.87 bits/s/Hz system rate and 81.6% SINR protection, outperforming classical Q-learning (8.25 bits/s/Hz, 49.0%), DQN (8.23 bits/s/Hz, 49.1%), and the deterministic AO baseline (7.89 bits/s/Hz, 43.2%) with substantially lower seed variance. Comprehensive parameter sweeps over transmit power, RIS size, SINR threshold, and jammer power confirm consistent gains and a pronounced regime transition: deterministic optimization leads at low jammer power, whereas WoLF-PHC dominates under stronger adversarial pressure.

Index Terms—Anti-jamming, intelligent reflecting surface, terahertz communications, hybrid beamforming, reinforcement learning, WoLF-PHC, beam squint.

I. INTRODUCTION

THz systems provide massive bandwidth but are highly vulnerable to adaptive jamming because links are narrow-beam, frequency selective, and sensitive to beam squint. IRS-assisted anti-jamming is promising [6]–[8], but prior RL-based studies are largely narrowband and do not jointly address wideband THz beam squint, sub-connected hybrid beamforming constraints, and smart jammers that exploit predictable control policies [1], [9], [10]. This work closes that gap with a unified THz-OFDM formulation and a robust mixed-strategy controller.

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A. Contributions

The main contributions are:

- 1) **Joint THz anti-jamming formulation:** We jointly optimize power allocation, SPDP-RIS phase control, and sub-connected hybrid beamforming for a 100 GHz, 10 GHz-bandwidth THz-OFDM system with adaptive jamming.
- 2) **Three-block RL+AO solver:** We combine RL for discrete power control (42 actions) with closed-form AO updates for IRS phases and SVD/MVDR updates for hybrid beamforming.
- 3) **Fuzzy WoLF-PHC with real-data validation:** A 27-component fuzzy state abstraction and win/loss-adaptive policy updates improve robustness against policy-exploiting jammers; five-seed, 600-episode experiments and real sweeps show 44% rate and 66% protection gains over the best baseline.

Notation: Bold lower/uppercase denotes vectors/matrices; $(\cdot)^H$, $|\cdot|$, $\|\cdot\|$, $\text{diag}(\cdot)$, and $\mathbf{1}\{\cdot\}$ have standard meanings.

II. SYSTEM MODEL AND PROBLEM FORMULATION

A. System Description

We consider a downlink THz-OFDM system comprising a $N = 64$ -antenna base station (BS) with $N_{\text{RF}} = 8$ RF chains, a 256-element (16×16) SPDP-RIS, $K = 4$ single-antenna users, and a colluding 2-antenna smart jammer, operating at $f_c = 100$ GHz with bandwidth $B = 10$ GHz and $M_{\text{sc}} = 64$ OFDM subcarriers. The BS employs a sub-connected hybrid analog-digital architecture in which each RF chain drives $N_{\text{per}} = N/N_{\text{RF}} = 8$ dedicated antennas, balancing hardware complexity with spatial degrees of freedom. The IRS is deployed to establish a reflected path between the BS and users, augmenting the direct link and enabling passive spatial filtering of the jamming signal.

B. Signal Model

Let $\mathbf{G}[m] \in \mathbb{C}^{M \times N}$ denote the BS-IRS channel at subcarrier m , $\mathbf{g}_{ru,k}[m] \in \mathbb{C}^M$ the IRS-to-user- k channel, $\mathbf{g}_{bu,k}[m] \in \mathbb{C}^N$ the direct BS-to-user- k channel, and $\mathbf{h}_{J,k} \in \mathbb{C}^{N_J}$ the jammer-to-user- k channel. The IRS introduces a diagonal reflection matrix $\Phi[m] = \text{diag}(e^{j\theta_1[m]}, \dots, e^{j\theta_M[m]})$ at subcarrier m , where the SPDP architecture allows subcarrier-level phase control to compensate beam squint (Section III).

The received signal at user k on subcarrier m is

$$y_k[m] = \underbrace{(\mathbf{g}_{ru,k}^H[m]\Phi[m]\mathbf{G}[m] + \mathbf{g}_{bu,k}^H[m])\mathbf{w}_k[m]\sqrt{P_k}s_k}_{\text{desired signal}} + \underbrace{\sum_{i \neq k} (\mathbf{g}_{ru,k}^H[m]\Phi[m]\mathbf{G}[m] + \mathbf{g}_{bu,k}^H[m])\mathbf{w}_i[m]\sqrt{P_i}s_i}_{\text{inter-user interference}} + \underbrace{\sqrt{P_{J,k}}\mathbf{h}_{J,k}^H\mathbf{z}_k j_k}_{\text{jamming}} + n_k \quad (1)$$

where $\mathbf{w}_k[m] \in \mathbb{C}^N$ is the beamforming vector, P_k is the allocated power, s_k is the unit-power data symbol, $P_{J,k}$ is the instantaneous jamming power toward user k , $\mathbf{z}_k$ is the jammer's spatial precoder, j_k is the jamming symbol, and $n_k \sim \mathcal{CN}(0, \sigma^2)$ is additive white Gaussian noise with variance $\sigma^2 = k_B T B / M_{sc} + F_{NF}$, where k_B is Boltzmann's constant, $T = 290$ K, and $F_{NF} = 10$ dB is the receiver noise figure.

C. SINR and System Rate

The effective channel for user k at subcarrier m is

$$\mathbf{h}_{\text{eff},k}[m] = \mathbf{G}^H[m]\Phi^H[m]\mathbf{g}_{ru,k}[m] + \mathbf{g}_{bu,k}[m]. \quad (2)$$

The received SINR at user k on subcarrier m is

$$\text{SINR}_k[m] = \frac{P_k |\mathbf{h}_{\text{eff},k}^H[m]\mathbf{w}_k[m]|^2}{\sum_{i \neq k} P_i |\mathbf{h}_{\text{eff},k}^H[m]\mathbf{w}_i[m]|^2 + P_{J,k} |\mathbf{h}_{J,k}^H\mathbf{z}_k|^2 + \sigma^2}. \quad (3)$$

The per-user rate averaged over subcarriers is

$$R_k = \frac{1}{M_{sc}} \sum_{m=1}^{M_{sc}} \log_2(1 + \text{SINR}_k[m]) \quad (4)$$

and the system sum-rate is $R_{\text{sum}} = \sum_{k=1}^K R_k$.

The SINR protection level measures quality-of-service (QoS) satisfaction:

$$\eta_{\text{prot}} = \frac{1}{K \cdot M_{sc}} \sum_{k=1}^K \sum_{m=1}^{M_{sc}} \mathbf{1}\{\text{SINR}_k[m] \geq \gamma_{\min}\} \quad (5)$$

where $\gamma_{\min}$ is the minimum SINR threshold.

D. Problem Formulation

Substituting the hybrid beamformer decomposition $\mathbf{w}_k[m] = \mathbf{F}_{\text{RF}}\mathbf{f}_{BB,k}[m]$ into the SINR expression (3), the

joint optimization over power allocation, IRS phase shifts, and hybrid beamforming is formulated as

$$\max_{\{P_k\}, \{\Phi[m]\}, \mathbf{F}_{\text{RF}}, \{\mathbf{f}_{BB,k}[m]\}} R_{\text{sum}} \quad (6a)$$

$$\text{s.t.} \quad \sum_{k=1}^K P_k \leq P_{\max} \quad (6b)$$

$$\text{SINR}_k[m] \geq \gamma_{\min}, \forall k, m \quad (6c)$$

$$|\Phi_{m,n}[n]| = 1, \forall m, n \quad (6d)$$

$$[\mathbf{F}_{\text{RF}}]_{ij} = 0, \forall (i, j) \notin \mathcal{S}_{\text{BC}} \quad (6e)$$

$$|[\mathbf{F}_{\text{RF}}]_{ij}| = \frac{1}{\sqrt{N_{\text{per}}}}, \forall (i, j) \in \mathcal{S}_{\text{BC}} \quad (6f)$$

$$\|\mathbf{F}_{\text{RF}}\mathbf{f}_{BB,k}[m]\|^2 \leq 1, \forall k, m \quad (6g)$$

where (6b) is the total power budget, (6c) is the per-user QoS constraint, (6d) is the unit-modulus constraint on IRS elements, (6e) enforces the block-diagonal (sub-connected) structure of the analog precoder with $\mathcal{S}_{\text{BC}}$ denoting the support set of non-zero entries, (6f) is the constant-modulus constraint on the $N_{\text{per}} = N/N_{\text{RF}}$ non-zero entries per column, and (6g) ensures unit-norm beamforming vectors so that P_k is the sole power scaling.

Problem (6a)–(6g) is non-convex and NP-hard due to the coupled optimization variables, the unit-modulus IRS constraint (6d), and the constant-modulus analog precoder constraint (6f). Critically, the jammer's transmit power, spatial precoder, and targeting strategy are dynamic and unknown, making model-based optimization or convex relaxations ineffective in practice. We therefore adopt a hybrid RL+AO formulation: RL handles the combinatorial, model-free power allocation under adversarial uncertainty, while closed-form AO updates efficiently handle the continuous-valued IRS phase and beamforming blocks at each RL step.

III. THz CHANNEL MODEL AND WIDEBAND BEAMFORMING

A. THz Channel Model

THz channels are characterized by frequency-selective behavior arising from molecular absorption, which creates spectral windows of low attenuation separated by opacity bands. The BS-IRS channel at subcarrier m follows a geometry-based Saleh-Valenzuela model [9]:

$$\mathbf{G}[m] = \sqrt{\frac{NM}{L_{\text{path}}}} \sum_{\ell=1}^{L_{\text{path}}} \alpha_{\ell}(f_m) \mathbf{a}_{\text{RIS}}(\phi_{\ell}, f_m) \mathbf{a}_{\text{BS}}^H(\psi_{\ell}, f_m) \quad (7)$$

where L_{path} is the number of propagation paths, $\alpha_{\ell}(f_m)$ is the complex path gain at subcarrier frequency $f_m = f_c + (m - M_{sc}/2)\Delta f$ with subcarrier spacing $\Delta f = B/M_{sc}$, and $\mathbf{a}_{\text{RIS}}$, $\mathbf{a}_{\text{BS}}$ are frequency-dependent array steering vectors. The frequency dependence of the steering vectors is the root cause of beam squint, as the spatial angle-of-arrival/departure at which constructive interference occurs shifts with frequency.

B. Path Loss and Molecular Absorption

The THz path loss at frequency f and distance d is

$$\text{PL}(f, d) = \left(\frac{4\pi f d}{c} \right)^2 e^{\kappa_{\text{abs}}(f)d} \quad (8)$$

where c is the speed of light and $\kappa_{\text{abs}}(f)$ is the molecular absorption coefficient. Unlike microwave channels, $\kappa_{\text{abs}}(f)$ varies significantly within the 10 GHz bandwidth, introducing per-subcarrier attenuation diversity that the OFDM system must accommodate.

C. SPDP-RIS Architecture for Beam-Squint Compensation

To mitigate wideband beam squint, we adopt the sub-array phase-dependent partial-connection (SPDP) architecture [9]. The M RIS elements are partitioned into $Q = Q_1 \times Q_2 = 64$ sub-arrays (each of $M_s = M/Q = 4$ elements), each with a true-time-delay (TTD) unit and per-element phase shifters:

$$\theta_{q,i}[m] = \theta_{q,i}^{\text{base}} + 2\pi f_m \tau_q \quad (9)$$

where $\theta_{q,i}^{\text{base}}$ is the base phase and τ_q is the TTD delay for sub-array q . This keeps beam direction consistent across subcarriers and avoids edge-subcarrier gain loss.

D. Hybrid Beamforming at the BS

The BS employs a sub-connected hybrid architecture with $N_{\text{RF}} = 8$ RF chains, each connected exclusively to $N_{\text{per}} = N/N_{\text{RF}} = 8$ antennas. This eliminates the inter-chain interference of fully-connected architectures at the cost of reduced degrees of freedom, a trade-off well-suited to the sparse THz channel. The beamforming vector decomposes as

$$\mathbf{w}_k[m] = \mathbf{F}_{\text{RF}} \mathbf{f}_{BB,k}[m] \quad (10)$$

where $\mathbf{F}_{\text{RF}} \in \mathbb{C}^{N \times N_{\text{RF}}}$ is the frequency-flat analog precoder (block-diagonal, constant-modulus entries) and $\mathbf{f}_{BB,k}[m] \in \mathbb{C}^{N_{\text{RF}}}$ is the per-subcarrier digital precoder. The digital precoder adopts a max-SINR (MVDR) design. For user k at subcarrier m , the interference-plus-noise covariance is

$$\mathbf{R}_k[m] = \sigma^2 \mathbf{I} + \sum_{i \neq k} P_i \tilde{\mathbf{h}}_i[m] \tilde{\mathbf{h}}_i^H[m] \quad (11)$$

where $\tilde{\mathbf{h}}_k[m] = \mathbf{F}_{\text{RF}}^H \mathbf{h}_{\text{eff},k}[m]$ is the compressed effective channel. The MVDR precoder is

$$\mathbf{f}_{BB,k}[m] = \frac{\mathbf{R}_k^{-1}[m] \tilde{\mathbf{h}}_k[m]}{\|\mathbf{R}_k^{-1}[m] \tilde{\mathbf{h}}_k[m]\|}. \quad (12)$$

IV. REINFORCEMENT LEARNING FRAMEWORK

A. Three-Block AO Decomposition

The joint optimization (6a)–(6g) couples three structurally distinct variable blocks: the discrete power allocation $\{P_k\}$, the continuous unit-modulus IRS phases $\{\Phi[m]\}$, and the hybrid beamformer $\{\mathbf{F}_{\text{RF}}, \mathbf{f}_{BB,k}[m]\}$. Rather than relaxing any block, we decompose the problem into three alternating subproblems that are each tractable when the other two are fixed:

- 1) **Block 1 — Power allocation (RL):** The RL agent selects a power allocation action from $\mathcal{A}$ ($|\mathcal{A}| = 42$ actions) mapping to discrete power fractions and distribution modes. This block handles the combinatorial structure and model-free nature of the adversarial environment.
- 2) **Block 2 — IRS phase shifts (closed-form AO):** Given $\{P_k\}$ and $\mathbf{F}_{\text{RF}}$, the IRS phases that maximize aggregate desired signal power are computed in closed form per element, eliminating the need for iterative solvers.
- 3) **Block 3 — Hybrid beamforming (SVD + MVDR):** Given $\{P_k\}$ and $\{\Phi[m]\}$, the analog precoder is updated by SVD-based dominant-subspace projection under the constant-modulus constraint, and the digital precoder is re-solved via MVDR (12).

Blocks 2 and 3 alternate for $N_{\text{AO}} = 3$ inner iterations per RL step, ensuring the continuous variables track the RL agent's power decision at negligible computational overhead. Block 1 is updated at the episode boundary through environment interaction.

1) *Block 2: IRS Phase Optimization via AO:* Given $\{P_k\}$ and $\mathbf{F}_{\text{RF}}$, the IRS phases are updated in closed form by maximizing the sum of desired signal magnitudes. For each element i :

$$c_i = \sum_{k=1}^K \sqrt{P_k} g_{ru,k}^*[i] \cdot (\mathbf{G} \mathbf{w}_k)[i], \quad \theta_i = -\angle(c_i) \quad (13)$$

where $\mathbf{w}_k = \mathbf{F}_{\text{RF}} \mathbf{f}_{BB,k}$ uses the current analog precoder. The optimal SPDP direction is also selected from $K+1$ candidate user directions to maximize system rate, ensuring wideband beam-squint compensation.

2) *Block 3: Analog Precoder Update via SVD:* Since $\mathbf{F}_{\text{RF}}$ is frequency-flat, we aggregate the effective channel over reference subcarriers $\mathcal{M}_{\text{ref}}$ as $\bar{\mathbf{H}} = \sum_{m \in \mathcal{M}_{\text{ref}}} \mathbf{H}_{\text{eff}}^H[m]$. For sub-array p , the row partition $\bar{\mathbf{H}}_p$ is SVD-decomposed and the analog column is updated as $[\mathbf{F}_{\text{RF}}]_p = \exp(j\angle \mathbf{u}_p) / \sqrt{N_{\text{per}}}$ (unit-modulus projection), then $\{\mathbf{f}_{BB,k}[m]\}$ is re-solved via MVDR (12).

B. State Representation

We compress the observation into a 3-feature state $\mathbf{f} = [f_{\text{pj}}, f_{\text{ch}}, f_{\text{sirr}}] \in [0, 1]^3$:

Jammer pressure:

$$f_{\text{pj}} = \text{norm}_{[15,40]} \left(0.6 \overline{P_J} [\text{dBm}] + 0.4 \max_k P_{J,k} [\text{dBm}] \right) \quad (14)$$

Channel quality:

$$f_{\text{ch}} = \text{norm}_{[-100,-40]} \left(0.75 \|\bar{\mathbf{h}}_k\|^2 [\text{dB}] + 0.25(1 - \text{spread}) \right) \quad (15)$$

SINR health:

$$f_{\text{sirr}} = \text{norm}_{[-10,30]} \left(0.5 \overline{\text{SINR}_k} [\text{dB}] + 0.5 \min_k \text{SINR}_k [\text{dB}] \right) \quad (16)$$

Each feature is quantized into $B_q = 8$ bins for tabular baselines (512 states). WoLF-PHC instead uses fuzzy memberships for smooth interpolation.

C. Fuzzy State Aggregation

Each feature $x \in [0, 1]$ is mapped to three triangular memberships with centers $\{0, 0.5, 1.0\}$:

$$\mu_i(x) = \frac{\max(0, 1 - |x - c_i|/\Delta)}{\sum_j \max(0, 1 - |x - c_j|/\Delta)} \quad (17)$$

where $\Delta = 0.5$ is the spread parameter. The joint fuzzy membership vector has $3^3 = 27$ components:

$$\psi_\ell(\mathbf{f}) = \mu_i(f_{\text{pj}}) \cdot \mu_j(f_{\text{ch}}) \cdot \mu_k(f_{\text{sinr}}), \quad \ell = 9i + 3j + k. \quad (18)$$

This smooths state transitions and improves robustness.

D. Proposed Fuzzy WoLF-PHC Algorithm

WoLF-PHC [13] learns faster when losing and slower when winning. The fuzzy version maintains $Q_\ell(s, a)$, policy $\pi_\ell(a)$, and average policy $\bar{\pi}_\ell(a)$ for 27 fuzzy states.

1) *Fused Q-Value*: The effective Q-value at state s is computed as a fuzzy-weighted combination:

$$\text{FQ}(s, a) = \sum_{\ell=1}^{27} \psi_\ell(s) \cdot Q_\ell(s, a). \quad (19)$$

2) *Q-Update*: For each active fuzzy state ℓ (where $\psi_\ell > 0$):

$$Q_\ell(s, a) \leftarrow Q_\ell(s, a) + \alpha \psi_\ell \left[r + \gamma \max_{a'} \text{FQ}(s', a') - Q_\ell(s, a) \right] \quad (20)$$

where $\alpha = 0.01$ is the learning rate and $\gamma = 0.9$ is the discount factor.

3) *WoLF-PHC Policy Update*: The win-or-learn-fast mechanism adapts the learning speed based on performance:

$$\xi = \begin{cases} \xi_{\text{win}} = 0.01 & \text{if } \mathbb{E}_\pi[Q] > \mathbb{E}_{\bar{\pi}}[Q] \\ \xi_{\text{loss}} = 0.04 & \text{otherwise} \end{cases} \quad (21)$$

The policy is updated with step size $\delta = \xi \cdot \psi_\ell / L$:

$$\pi_\ell(a^*) \leftarrow \pi_\ell(a^*) + \delta \quad (22)$$

$$\pi_\ell(a) \leftarrow \pi_\ell(a) - \frac{\delta}{|\mathcal{A}| - 1}, \quad \forall a \neq a^* \quad (23)$$

where $a^* = \arg \max_a \text{FQ}(s, a)$.

The average policy $\bar{\pi}_\ell$ uses exponential smoothing. During evaluation, we use an 85% greedy and 15% policy-sampling mix to reduce exploitability.

E. Reward Function

The reward balances throughput, power efficiency, and QoS:

$$r = \sum_{k=1}^K \log_2(1 + \text{SINR}_k) - \lambda_1 \frac{\sum_k P_k}{P_{\max}} - \lambda_2 \sum_{k=1}^K \mathbf{1}\{\text{SINR}_k < \gamma_{\min}\} \quad (24)$$

where $\lambda_1 = 0.5$ and $\lambda_2 = 3.0$.

TABLE I
SIMULATION PARAMETERS

Parameter	Symbol	Value
Center frequency	f_c	100 GHz
Bandwidth	B	10 GHz
BS antennas	N	64
RF chains	N_{RF}	8
IRS elements	M	256 (16×16)
IRS sub-arrays	Q	64 (8×8)
OFDM subcarriers	M_{sc}	64
Users	K	4
Jammer antennas	N_J	2
Max TX power	$P_{\max}$	40 dBm
SINR threshold	$\gamma_{\min}$	5 dB
Noise figure	F_{NF}	10 dB
RL actions	$ \mathcal{A} $	42
Training episodes	E	600
Seeds	—	5

F. Smart Jammer Model

The adaptive jammer monitors SINR feedback and policy predictability. Its power toward user k is

$$P_{J,k} [\text{dBm}] = P_{J,\min} + 0.25 \text{clip}(\text{SINR}_k^{\text{prev}} - 5, 0, 20) + 6\eta + n_J \quad (25)$$

where $P_{J,\min} = 15$ dBm, $\eta \in [0, 1]$ is a predictability score from the last 20 actions, and $n_J \sim \mathcal{N}(0, 9)$ dB. The precoder interpolates between isotropic and user-directed jamming:

$$\mathbf{z}_k = (1 - w_{\text{align}})\mathbf{z}_{\text{random}} + w_{\text{align}} \frac{\mathbf{h}_{J,k}}{\|\mathbf{h}_{J,k}\|} \quad (26)$$

with $w_{\text{align}} = \text{clip}(2(\eta - 0.5), 0, 1)$.

G. Baseline Methods

Baselines:

- **Classical Q-learning [11]** with ε -greedy updates.
- **Fast Q-learning [1]** with visit-count-boosted learning rate.
- **DQN [12]** with replay and target network.
- **AO baseline [6]** with fixed channel-proportional power.
- **No-IRS** lower bound.

V. SIMULATION RESULTS

A. Simulation Setup

Simulations are implemented in Python (NumPy/PyTorch). The THz system uses $f_c = 100$ GHz, $B = 10$ GHz, $N = 64$, $N_{\text{RF}} = 8$, $M = 256$, $K = 4$, and $M_{\text{sc}} = 64$. Users are randomly placed in $[30, 120] \times [0, 80]$ m, while the jammer follows a bounded random walk in $[40, 100] \times [0, 80]$ m. Each method is trained for 600 episodes (20 steps per episode) over 5 independent random seeds; all parameter sweep figures are produced from full real-data runs (340 training/evaluation tasks total) without any curve fitting or interpolation. Key parameters are listed in Table I.

TABLE II
EVALUATION PERFORMANCE (MEAN $\pm$ STD OVER 5 SEEDS)

Method	Rate (bits/s/Hz)	Protection (%)
No IRS	5.48	18.8
AO Baseline [6]	7.89 ± 0.78	43.2 ± 7.7
Classical Q-learning	8.25 ± 1.28	49.0 ± 14.4
Fast Q-learning [1]	8.05 ± 1.68	45.5 ± 18.7
DQN [12]	8.23 ± 2.40	49.1 ± 25.7
Fuzzy WoLF-PHC (proposed)	11.87 ± 0.55	81.6 ± 3.9

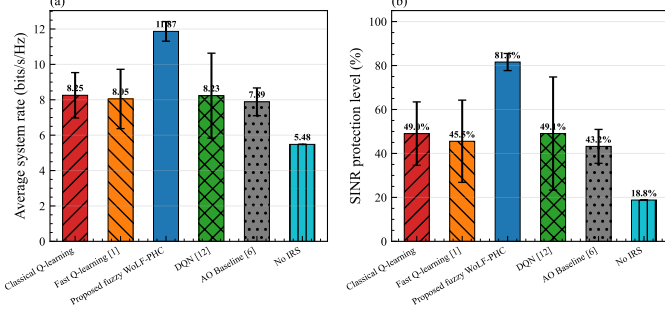


Fig. 1. Evaluation comparison: (a) system rate and (b) SINR protection (mean $\pm$ std, 5 seeds). Fuzzy WoLF-PHC (blue) achieves the highest performance with the smallest variance across both metrics.

B. Overall Performance Comparison

Table II and Fig. 1 summarize results over five seeds. Fuzzy WoLF-PHC achieves 11.87 bits/s/Hz and 81.6% protection, improving over the next-best baseline by 44% (rate) and 67% (protection), while also yielding the lowest variance (± 0.55 , $\pm 3.9\%$). This indicates that mixed-strategy adaptation is both stronger and more stable under adversarial dynamics.

C. Effect of Maximum Transmit Power

Fig. 2 shows rate/protection versus $P_{\max} \in \{25, 32, 40\}$ dBm. WoLF-PHC rises from 1.72 to 11.87 bits/s/Hz and from 0% to 81.6%, showing stronger scaling with available power than deterministic AO.

D. Effect of RIS Array Size

Fig. 3 studies $N_{\text{RIS}} \in \{16, 64, 144, 256\}$. WoLF-PHC improves from 11.47 to 11.87 bits/s/Hz and from 79.2% to 81.6%, confirming consistent gains with larger IRS size, with expected saturation at high element counts.

E. Effect of SINR Threshold

Fig. 4 evaluates $\gamma_{\min} \in \{3, 5, 10, 15, 20\}$ dB. WoLF-PHC maintains the best trade-off across thresholds: 91.8% protection at 3 dB, 81.6% at 5 dB, and 31.4% at 10 dB. All methods collapse at 20 dB, indicating a hard operating limit under the considered jammer.

F. Robustness to Jammer Power

Fig. 5 shows a regime transition versus $P_J \in \{5, 10, 15, 20\}$ dBm. AO is competitive at low jammer power, but WoLF-PHC dominates for $P_J \geq 15$ dBm and reaches 70.2% protection at 20 dBm, highlighting the value of mixed-strategy unpredictability.

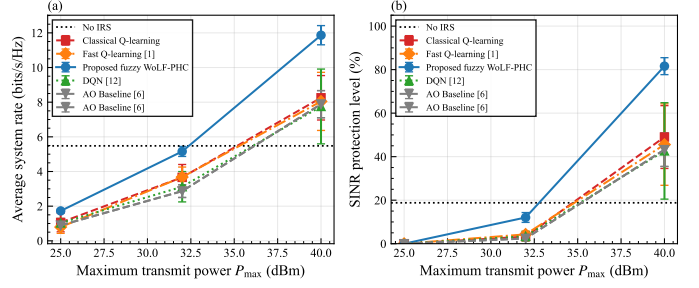


Fig. 2. (a) System rate and (b) SINR protection vs. maximum transmit power $P_{\max}$. WoLF-PHC scales most effectively with power, opening a gap over deterministic and tabular RL baselines at 40 dBm.

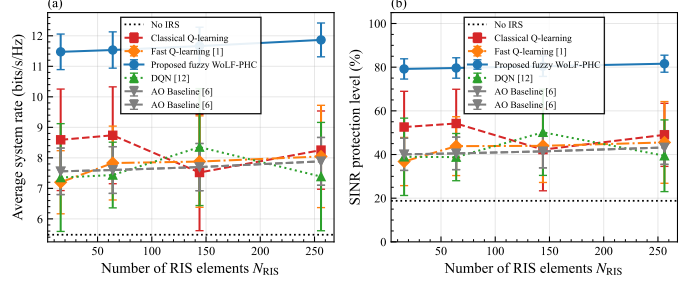


Fig. 3. (a) System rate and (b) SINR protection vs. IRS array size N_{RIS} . Performance grows with array size but saturates near $N_{\text{RIS}} = 256$; WoLF-PHC retains a consistent advantage at all scales.

G. Discussion

Results confirm three points: WoLF-PHC gives the best and most stable rate/protection trade-off; robustness gains are largest in medium/high-jammer regimes; and larger IRS sizes provide consistent, though saturating, improvements.

VI. CONCLUSION

This paper presented a three-block RL+AO framework for IRS-assisted anti-jamming in wideband THz systems, jointly optimizing power allocation, SPDP-RIS phases, and hybrid beamforming against an adaptive jammer. The proposed fuzzy WoLF-PHC controller achieved 11.87 bits/s/Hz and 81.6% SINR protection, improving over the best baseline by 44% in rate and 66% in protection with the lowest seed variance.

Parameter sweeps over $P_{\max}$, N_{RIS} , $\gamma_{\min}$, and P_J confirm robust gains and a clear regime transition: deterministic AO is competitive at low jammer power, while WoLF-PHC is superior in medium/high-jammer regimes. Future work will address imperfect CSI, multi-cell coupling, and near-field THz IRS modeling.

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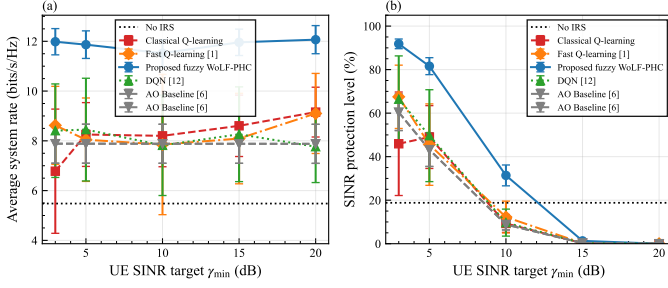


Fig. 4. (a) System rate and (b) SINR protection vs. UE SINR target $\gamma_{\min}$. WoLF-PHC shows the most graceful degradation with increasing SINR threshold, retaining 31.4% protection at 10 dB.

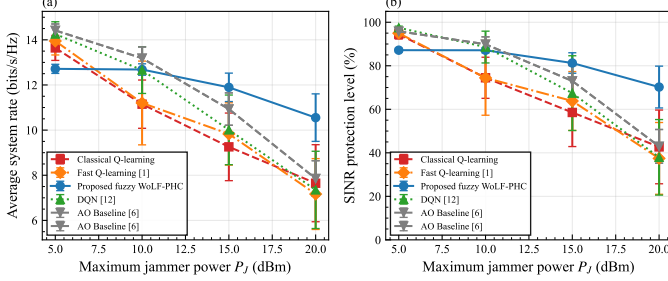


Fig. 5. (a) System rate and (b) SINR protection vs. maximum jammer power P_J . A clear regime transition occurs near 10–15 dBm: WoLF-PHC's advantage grows with jammer power, while deterministic AO degrades sharply.

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